# IGCSE 0607 Practice Questions — E7.6 — Combined Questions & Markscheme

## Pythagoras' theorem and trigonometry in 3D

**Total Questions:** 3 — **Total Marks:** 19 Marks

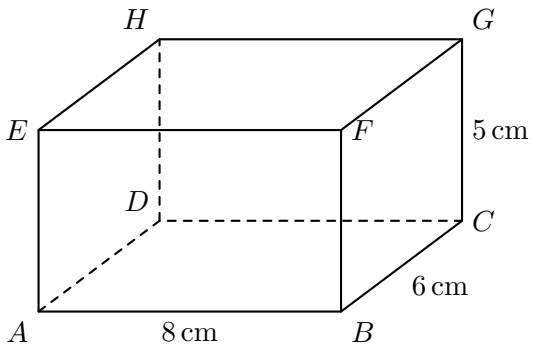
---

## Table of Contents & Quick Navigation


- **E7.6** Pythagoras' theorem and trigonometry in 3D
  → **Paper 4 (Calculator)** *(3 Qs)*


---

**1** The diagram shows a cuboid $ABCDEFGH$.
$AB = 8$ cm, $BC = 6$ cm and $CG = 5$ cm.




**(a)** Calculate the length of the diagonal $AC$.

..... cm **[2]**

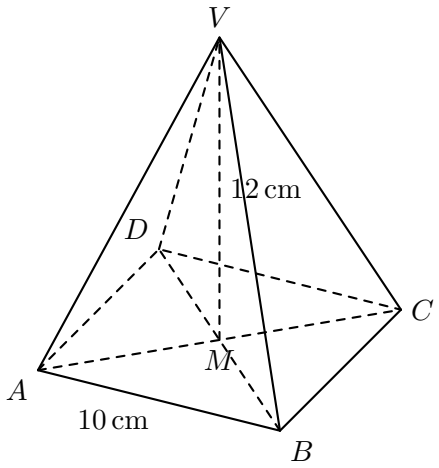
**(b)** Calculate the length of the space diagonal $AG$.

..... cm **[2]**

**(c)** Calculate the angle between the line $AG$ and the base $ABCD$.

..... ° **[2]**

**2** The diagram shows a pyramid $VABCD$ with a square base $ABCD$ of side 10 cm.
The point $M$ is the centre of the base and $V$ is vertically above $M$.
The vertical height $VM$ of the pyramid is 12 cm.



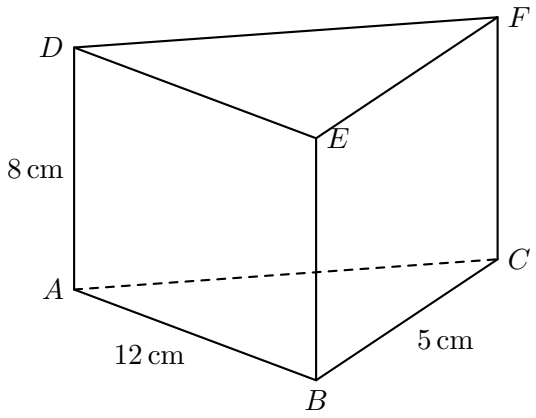

**(a)** Calculate the length of the edge $VA$.

..... cm **[3]**

**(b)** Calculate the angle between the edge $VA$ and the base $ABCD$.

..... ° **[2]**

**3** The diagram shows a right triangular prism $ABCDEF$.
The base $ABC$ is a horizontal right-angled triangle with angle $ABC = 90°$.
$AB = 12$ cm, $BC = 5$ cm and the vertical height $AD = 8$ cm.




**(a)** Calculate the length of $AF$.

..... cm **[3]**

**(b)** Calculate the angle between the line $AF$ and the base $ABC$.

..... ° **[2]**

**(c)** Calculate the angle between the line $AF$ and the vertical plane $ABED$.

..... ° **[3]**

**1** **(a)** $AC^2 = 8^2 + 6^2$ **[M1]**

$AC = 10$ **[A1]**

**(b)** $AG^2 = 10^2 + 5^2$ (or $AG^2 = 8^2 + 6^2 + 5^2$) **[M1]**

$AG = \sqrt{125} = 11.18...$ so 11.2 **[A1]**

**(c)** $\tan\theta = \dfrac{5}{10}$ (or $\sin\theta = \dfrac{5}{\sqrt{125}}$, etc.) **[M1]**

$\theta = 26.56...$ so $26.6°$ **[A1]**

[Back to Question] *[Total Marks: 6]*

**2** **(a)** Uses Pythagoras to find half diagonal $AM$: $AM = \dfrac{1}{2}\sqrt{10^2 + 10^2} = \sqrt{50}$ (or 7.07...) **[M1]**

$VA^2 = 12^2 + (\sqrt{50})^2 = 144 + 50 = 194$ **[M1]**

$VA = \sqrt{194} = 13.92...$ so 13.9 **[A1]**

**(b)** $\tan(\angle VAM) = \dfrac{12}{\sqrt{50}}$ **[M1]**

$\angle VAM = 59.49...$ so $59.5°$ **[A1]**

[Back to Question] *[Total Marks: 5]*

**3** **(a)** Uses Pythagoras to find $AC$: $AC = \sqrt{12^2 + 5^2} = 13$ **[M1]**

$AF^2 = 13^2 + 8^2 = 169 + 64 = 233$ **[M1]**

$AF = \sqrt{233} = 15.26...$ so 15.3 **[A1]**

**(b)** Identifies required angle as $\angle FAC$ and uses trig, e.g. $\tan(\angle FAC) = \dfrac{8}{13}$ **[M1]**

$\angle FAC = 31.60...$ so $31.6°$ **[A1]**

**(c)** Recognises the required angle is $\angle FAE$ because $FE$ is perpendicular to plane $ABED$ **[B1]**

Uses trig in triangle $FAE$ (where $\angle FEA = 90°$): $\sin(\angle FAE) = \dfrac{5}{\sqrt{233}}$ OR finds

$AE = \sqrt{12^2 + 8^2} = \sqrt{208}$ and uses $\tan(\angle FAE) = \dfrac{5}{\sqrt{208}}$ **[M1]**

$\angle FAE = 19.11...$ so $19.1°$ **[A1]**

[Back to Question] *[Total Marks: 8]*